

Alignment

You can run an alignment while logged out, but you must be logged in to **Save** it and later reopen it under **Manage alignments** (see the dedicated guidelines for accounts). This walkthrough uses Gradual and Tract datasets (see the dedicated data management guidelines for instructions on how to create such datasets).

This guideline first show how to align **Tracts** by pitches: colours and display modes, saving and downloads, the guide tree, the distance matrix, and a snapshot. It then runs a concatenated pitch alignment on **Graduals** with phylogenetic analysis. In the end we show how to use **Manage alignments** tab (reopen, delete, or upload a saved JSON).

Tracts

1. Pick Tracts and select all

The chant list only shows sources that are checked in **Data sources**.

- 1. Open the **Data sources** panel if it is collapsed.
- 2. Uncheck the shared default corpora.
- 3. Check **Tracts**.
- 4. Click **Save**.

Home

Chant list

Manage data

Manage alignments

Dashboard

Settings

About

docsguide882194

Logout

Search incipit

☐	[...] munt caelestis gloriae	a02376	CDN-Mlr MS Medieval 0073, 087r, M	Antiphon	First Vespers	7	CantusCorpus v1.0
☐	[...] o madens ut flumine	a02053	CDN-Mlr MS Medieval 0073, 010r, 3.1	Responsory	Matins	8	CantusCo v1.0
☐	[Ad te domine levavi animam	g00698a	CDN-Mlr MS Medieval 0092, 002, 01	Tract verse	Unknown	2	CantusCo v1.0
☐	[Adorna thalamum tuum Sion et	g00068	FIN-Hy F.m.1.85, 017r, ?	Antiphon	Others	6T	CantusCo v1.0
☐	[Alleluia Angelus domini descendit de	008415.2	FIN-Hy F.m.1.26, 011r,	Alleluia	Mass	8T	CantusCo v1.0
☐	[Alleluia Benedictus es] domine deus	g01113	FIN-Hy F.m.1.85, 013rb,	Alleluia	Mass	5?	CantusCorpus v1.0
☐	[Alleluia Confitemini domino quoniam bonus	507011	FIN-Hy F.m.1.26, 009v,	Alleluia	Mass	?	CantusCorpus v1.0
☐	[Alleluia Domine] refugium factus es	g01197	CDN-Mlr MS Medieval 0007, 060r,	Alleluia	Mass	7	CantusCorpus v1.0
☐	[Alleluia Posuisti] domine	g01302	CDN-LonONpc D-01oko, 075r,	Alleluia	Mass	1	CantusCorpus v1.0
☐	[Alma redemptoris mater quae pervia	001356	CDN-Mlr MS Medieval 0170, 291,	Antiphon	Others	5	CantusCorpus v1.0
☐	[Ave gratia plena dei genetrix	200456	FIN-Hy F.m.1.26, 020v,	Antiphon	Mass	1	CantusCorpus v1.0
☐	[Beatus dei athleta Vincentius horrendo	006201	CDN-WAdl MS Medieval 01, M6364r, 3.1	Responsory	Matins	1	CantusCorpus v1.0
☐	[Benedicite gentes dominum deum nostrum	g01070	FIN-Hy F.m.1.26, 005r,	Offertory	Unknown	2	CantusCorpus v1.0
☐	[Benedictus sis domine qui] tanta	200672	A-KN CCl 1018, 262r, B	Antiphon	Lauds	5	CantusCorpus v1.0
☐	[Cenantibus autem accepit] Jesus panem	001781	CDN-Mlr MS Medieval 0025, 071r, M	Antiphon	First Vespers	1	CantusCorpus v1.0
☐	[Christus resurgens ex mortuis] jam	g01031	FIN-Hy F.m.1.26, 012v,	Communion	Mass	8	CantusCorpus v1.0

Items per page: 20

1 - 20 of 53097

< >

Export to CSV

Create Dataset

Add to dataset

Data sources

☐ CantusCorpus v1.0

☐ netvor-0.3

☐ Graduals

☒ Tracts

Save

Confirm **Source selection saved**. The table now contains only chants from that dataset.

Click **Select/deselect all**. That ticks every chant in the dataset, not only the rows on the current page. The **(Selected: ...)** counter updates.

[Home](#)
[Chant list](#)
[Manage data](#)
[Manage alignments](#)
[Dashboard](#)
[Settings](#)
[About](#)
[docsguide882194](#)
[Logout](#)

Filter search

Apply filters and refresh

Align word-based

Align pitches

Align intervals

☒ Select/deselect all (Selected: 38)
 ☐ Sort by Cantus ID frequency
 ☐ Concatenated alignment
 ☐ Keep liquescents

Incipit	Cantus ID	Source	Genre	Office	Mode	Dataset name
☒ Absolve domine animas omnium fidelium	g01569	US-Stu MSS CODEX 0734 FF, 080v,	Tract	Unknown	8	Tracts
☒ Ad te levavi oculos meos	g00747	CH-P 18, 108,	Tract	Unknown	8	Tracts
☒ Attende caelum et loquar et	g02373	CH-P 18, 228, 1	Tract	Unknown	8	Tracts
☒ Audi filia et vide et	g01384	CH-P 18, 287,	Tract	Unknown	2	Tracts
☒ Ave Maria gratia plena dominus	g00119	CH-P 18, 296,	Tract	Unknown	2	Tracts
☒ Beatus vir qui timet dominum	g01285	CH-P 18, 295,	Tract	Unknown	8	Tracts
☒ Cantemus domino gloriose enim honorificatus	g00984	CH-P 18, 193,	Tract	Mass	8	Tracts
☒ Commovisti domine terram et conturbasti	g00643	CH-P 18, 060,	Tract	Unknown	8	Tracts
☒ Confitemini domino quoniam bonus quoniam	g00722	CH-P 18, 094,	Tract	Unknown	2	Tracts
☒ De necessitatibus meis eripe me	g00698	CH-P 18, 083,	Tract	Unknown	2	Tracts
☒ De profundis clamavi ad te	g00634	CH-P 18, 057,	Tract	Unknown	8	Tracts
☒ Desiderium animae ejus tribuisti ei	g01275	CH-P 18, 290,	Tract	Unknown	8	Tracts
☒ Deus deus meus respice in	g00848	CH-P 18, 161,	Tract	Unknown	2	Tracts
☒ Diffusa est gratia in labiis	g02214	CH-P 18, 286,	Tract	Unknown	2	Tracts
☒ Domine audivi auditum tuum et	g02429	CDN-Mlr MS Medieval 0025, 071r, 1	Tract	Unknown	2	Tracts

2. Align pitches

Leave **Concatenated alignment** unchecked. Click **Align pitches**. The other two buttons are:

- **Align word-based** — syllable / word-aware alignment
- **Align intervals** — alignment on interval sequences instead of pitches

Home

Chant list

Manage data

Manage alignments

Dashboard

Settings

About

docsguide882194

Logout

Search incipit

Filter search

Apply filters and refresh

Align word-based

Align pitches

Align intervals

Select/deselect all

(Selected: 38)

Sort by Cantus ID frequency

Concatenated alignment

Keep liquescents

	Incipit	Cantus ID	Source	Genre	Office	Mode	Dataset name
	Absolve domine animas omnium fidelium	g01569	US-STu MSS CODEX 0734 FF, 080v,	Tract	Unknown	8	Tracts
	Ad te levavi oculos meos	g00747	CH-P 18, 108,	Tract	Unknown	8	Tracts
	Attende caelum et loquar et	g02373	CH-P 18, 228, 1	Tract	Unknown	8	Tracts
	Audi filia et vide et	g01384	CH-P 18, 287,	Tract	Unknown	2	Tracts
	Ave Maria gratia plena dominus	g00119	CH-P 18, 296,	Tract	Unknown	2	Tracts
	Beatus vir qui timet dominum	g01285	CH-P 18, 295,	Tract	Unknown	8	Tracts
	Cantemus domino gloriose enim honorificatus	g00984	CH-P 18, 193,	Tract	Mass	8	Tracts
	Commovisti domine terram et conturbasti	g00643	CH-P 18, 060,	Tract	Unknown	8	Tracts
	Confitemini domino quoniam bonus quoniam	g00722	CH-P 18, 094,	Tract	Unknown	2	Tracts
	De necessitatibus meis eripe me	g00698	CH-P 18, 083,	Tract	Unknown	2	Tracts
	De profundis clamavi ad te	g00634	CH-P 18, 057,	Tract	Unknown	8	Tracts
	Desiderium animae ejus tribuisti ei	g01275	CH-P 18, 290,	Tract	Unknown	8	Tracts
	Deus deus meus respice in	g00848	CH-P 18, 161,	Tract	Unknown	2	Tracts
	Diffusa est gratia in labiis	g02214	CH-P 18, 286,	Tract	Unknown	2	Tracts
	Domine audiavi auditum tuum et	g02429	CDN-Mlr MS Medieval 0025, 071r, 1	Tract	Unknown	2	Tracts

Data sources

Wait until the alignment page is ready.

If a warning lists chants whose text and melody do not match, click **Dismiss**. Those items may stay in the alignment with incomplete text underlay, or be omitted.

Home

Chant list

Manage data

Manage alignments

Dashboard

Settings

About

docsguide882194

Logout

Search incipit

Alignment warning. The chant text and the melody do not have the same number of words or syllables, so they cannot be underlaid correctly. 1 included with incomplete text underlay. All selected chants remain in the alignment.

INCLUDED IN THE ALIGNMENT

- Veni sponsa [Christi accipe coronam

Display options

Display mode

Save

Download Alignment JSON

Download FASTA

Download Newick

Show headers

Volpiano

Raw

Distance Matrix

Download Distance Matrix

Guide Tree

Each block is one aligned sequence. The title is a number plus a short name built from incipit, source and a mode. Click the title to open chant details.

[Home](#)
[Chant list](#)
[Manage data](#)
[Manage alignments](#)
[Dashboard](#)
[Settings](#)
[About](#)

docsguide882194

[Logout](#)

Search incipit

Display options

☒ Show headers
 ☒ Show text
 ☐ Show colors
 ☐ Show conservation
 ☐ Segment words

Display mode

☒ Volpiano
 ☐ Raw

38 sequences

☒ Notes
 ☐ All

Save

Download Alignment JSON

Download FASTA

Download Newick

Distance Matrix

Download Distance Matrix

Guide Tree

Download Alignment Snapshot PNG

Reduce to Contrafacts

Phylogenetic Analysis

1: Absolve domine animas omnium fidelium US-STu MSS CODEX 0734 FF 778051 mode 8

2: Attende caelum et loquar et CH-P 18 777856 mode 8

3: Vinea facta est dilecto in CH-P 18 778818 mode 8

4: Cantemus domino gloriose enim honorificatus CH-P 18 390637 mode 8

3. Colours and display modes

The left **Display options** and **Display mode** controls change how the same alignment is drawn. They do not recompute it.

Check **Show colors**. Matching pitches across sequences share a colour, so conserved notes and gaps stand out. **Show conservation** is an alternative colouring (the two checkboxes replace each other).

☒ Show colors
 ☐ Show conservation
 ☐ Segment words

☒ Notes
 ☐ All

Phylogenetic Analysis

1: Absolve domine animas omnium fidelium US-STu MSS CODEX 0734 FF 778051 mode 8

2: Attende caelum et loquar et CH-P 18 777856 mode 8

3: Vinea facta est dilecto in CH-P 18 778818 mode 8

4: Cantemus domino gloriose enim honorificatus CH-P 18 390637 mode 8

Show headers, **Show text**, and **Segment words** toggle names, lyrics, and word bars.

With display mode **Volpiano**, a second toggle appears:

- **Notes** — only note characters (the default)
- **All** — also bars, clefs, and other Volpiano punctuation

Switch **Display mode** to **Raw** to see the aligned strings in a monospace encoding instead of the Volpiano font.

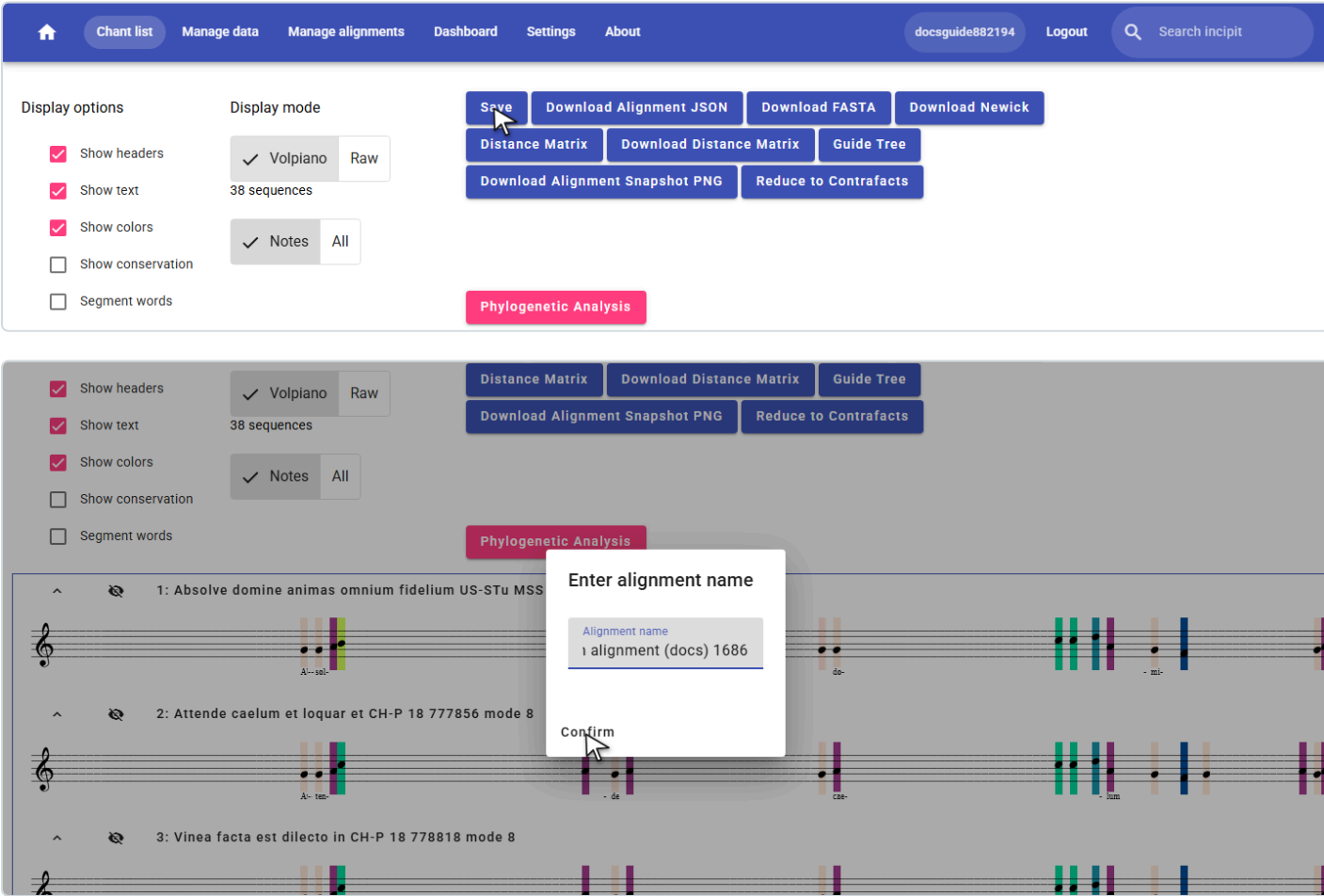
The screenshot shows the Incipit web application interface. The top navigation bar includes links for Home, Chant list, Manage data, Manage alignments, Dashboard, Settings, and About. A user profile 'docsguide882194' and a Logout button are on the right, along with a search bar labeled 'Search incipit'. The 'Display options' section on the left has checkboxes for 'Show headers', 'Show text', 'Show colors', 'Show conservation', and 'Segment words', all of which are checked. The 'Display mode' section shows 'Volpiano' selected, with a 'Raw' toggle also visible. Below these are buttons for 'Save', 'Download Alignment JSON', 'Download FASTA', 'Download Newick', 'Distance Matrix', 'Download Distance Matrix', 'Guide Tree', 'Download Alignment Snapshot PNG', and 'Reduce to Contrafacts'. A 'Phylogenetic Analysis' button is also present. The main content area displays four chant sequences, each with a title and mode 8. The sequences are: 1: Absolve domine animas omnium fidelium US-STu MSS CODEX 0734 FF 778051 mode 8; 2: Attende caelum et loquar et CH-P 18 777856 mode 8; 3: Vinea facta est dilecto in CH-P 18 778818 mode 8; 4: Cantemus domino gloriose enim honorificatus CH-P 18 390637 mode 8. Each sequence is shown in a monospace encoding, with notes and punctuation represented by characters like 'g', 'a', 'c', 'e', 'l', 'u', 'm', 'e', 't', 'l', 'o', 'q', 'u', 'a', 'r', 'e', 't', 'c', 'h', 'p', '18', '777856', 'm', 'o', 'd', 'e', '8'.

Switch back to **Volpiano** and **Notes** for the usual melody layout.

The screenshot shows the Incipit web application interface with the 'Notes' display mode selected. The top navigation bar and 'Display options' section are the same as in the previous screenshot. The 'Display mode' section now shows 'Notes' selected, with 'Volpiano' and 'All' toggles also visible. The main content area displays the same four chant sequences, but now they are shown in a standard musical notation layout. Each sequence is shown on a five-line staff with a treble clef. The notes are represented by black dots on the lines, and the punctuation is represented by vertical bars. The sequences are: 1: Absolve domine animas omnium fidelium US-STu MSS CODEX 0734 FF 778051 mode 8; 2: Attende caelum et loquar et CH-P 18 777856 mode 8; 3: Vinea facta est dilecto in CH-P 18 778818 mode 8; 4: Cantemus domino gloriose enim honorificatus CH-P 18 390637 mode 8.

4. Save and download

Save stores the alignment on your account so you can reopen it under **Manage alignments**. Enter a name and **Confirm**.



The functionality of other buttons next to the **Save** button:

Button	What it does
Download Alignment JSON	Full ChantLab alignment (can be uploaded again under Manage alignments)
Download FASTA	Aligned sequences in FASTA
Download Newick	Guide-tree Newick
Download Distance Matrix	Pairwise distances as a file
Download Alignment Snapshot PNG	Picture of the visible alignment (JPEG or PDF if that format is selected under Settings)
Reduce to Contrafacts	Keeps only sequences that look like contrafacts of each other (hidden in concatenated mode)

JSON is the format to archive or share a ChantLab alignment. FASTA and Newick are for external tools.

5. Guide tree

Click **Guide Tree**. This is the similarity tree MAFFT used while aligning. Drag the card if it covers the sequences. Click **Guide Tree** again to hide it. Word-based alignment does not offer a guide tree.

Chant list

Manage data

Manage alignments

Dashboard

Settings

About

docsguide882194

Logout

Search incipit

Display options

Show headers

Show text

Show colors

Show conservation

Segment words

Display mode

Volpiano

Raw

38 sequences

Save

Download Alignment JSON

Download FASTA

Download Newick

Distance Matrix

Download Distance Matrix

Guide Tree

Download Alignment Snapshot PNG

Reduce to Contrafacts

Guide tree

6. Distance matrix

Click **Distance Matrix**. Cells are pairwise distances between the visible sequences. Click **Distance Matrix** again to hide it.

Chant list

Manage data

Manage alignments

Dashboard

Settings

About

docsguide882194

Logout

Search incipit

Display options

Show headers

Show text

Display mode

Volpiano

Raw

38 sequences

Save

Download Alignment JSON

Download FASTA

Download Newick

Distance Matrix

Download Distance Matrix

Guide Tree

Download Alignment Snapshot PNG

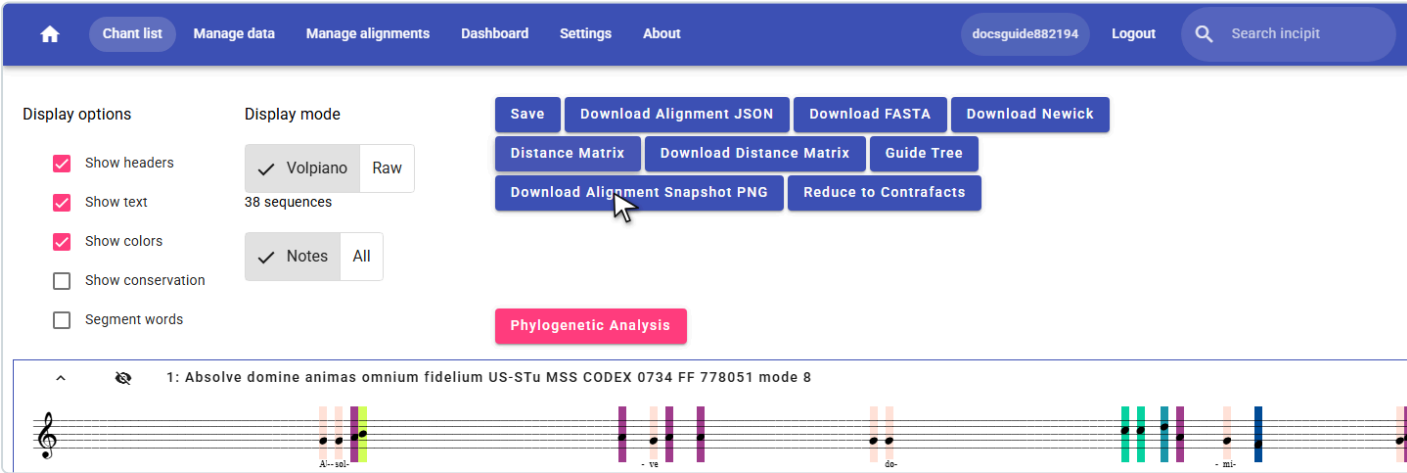
Reduce to Contrafacts

Distance matrix

	Absolve domine	Attende caelum	Vinea facta	Cantemus domino	Sicut cervus	Qui confidunt	Beatus vir	Jubilate domino	Laudate dominum	Laudate dominum	De profundis	Qui confidunt	Qui regis	Qui seminant	Qui seminant	Saepa expugnauerunt	Ad te	Commovisti domine	Desiderium animae	Veni sponsa	Quasi stella	Audi filia	Diffusa est	Eripe me	Eripe me	Qui habitat	Tu es	Deus deus	Confitemini domino	Gaude Maria	Ave Maria	Domine audivi	Domine audivi	Domine exaudi	Domine exaudi	Domine non	Domine non	De necessitatibus
Absolve domine	0.00	0.35	0.27	0.54	0.34	0.50	0.58	0.63	0.52	0.52	0.62	0.69	0.69	0.82	0.82	0.83	0.76	0.75	0.79	0.98	0.83	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.96	0.97	0.96	0.96	0.97	0.98	0.95	
Attende caelum	0.35	0.00	0.14	0.52	0.23	0.56	0.54	0.60	0.49	0.50	0.61	0.69	0.70	0.81	0.81	0.85	0.74	0.73	0.79	0.98	0.86	0.97	0.96	0.96	0.97	0.95	0.96	0.97	0.96	0.97	0.97	0.97	0.96	0.96	0.97	0.98	0.95	
Vinea facta	0.27	0.14	0.00	0.54	0.24	0.53	0.49	0.55	0.43	0.44	0.56	0.68	0.66	0.79	0.79	0.82	0.72	0.71	0.78	0.97	0.83	0.96	0.97	0.97	0.97	0.96	0.96	0.96	0.96	0.97	0.97	0.96	0.96	0.97	0.98	0.95		
Cantemus domino	0.54	0.52	0.54	0.00	0.45	0.38	0.71	0.74	0.67	0.68	0.74	0.76	0.78	0.87	0.87	0.89	0.84	0.84	0.84	0.99	0.90	0.98	0.98	0.98	0.98	0.97	0.97	0.98	0.97	0.98	0.98	0.97	0.98	0.97	0.98	0.96		
Sicut cervus	0.34	0.23	0.24	0.45	0.00	0.51	0.56	0.61	0.50	0.51	0.59	0.61	0.63	0.78	0.79	0.82	0.76	0.75	0.77	0.98	0.85	0.97	0.97	0.97	0.97	0.96	0.96	0.97	0.96	0.97	0.96	0.97	0.96	0.96	0.97	0.98	0.95	
Qui confidunt	0.50	0.56	0.53	0.38	0.51	0.00	0.69	0.73	0.65	0.66	0.73	0.75	0.77	0.86	0.86	0.89	0.83	0.82	0.82	0.99	0.90	0.98	0.97	0.97	0.98	0.97	0.97	0.97	0.98	0.98	0.98	0.96	0.97	0.97	0.98	0.98	0.96	
Beatus vir	0.58	0.54	0.49	0.71	0.56	0.69	0.00	0.35	0.28	0.28	0.38	0.66	0.60	0.73	0.72	0.73	0.74	0.73	0.78	0.98	0.86	0.96	0.97	0.96	0.96	0.96	0.96	0.96	0.96	0.97	0.96	0.96	0.96	0.96	0.97	0.97	0.94	
Jubilate domino	0.63	0.60	0.55	0.74	0.61	0.73	0.35	0.00	0.27	0.18	0.37	0.69	0.60	0.79	0.79	0.79	0.75	0.75	0.80	0.98	0.84	0.98	0.98	0.98	0.98	0.97	0.98	0.97	0.99	0.98	0.97	0.97	0.97	0.98	0.99	0.95		
Laudate dominum	0.52	0.49	0.43	0.67	0.50	0.65	0.28	0.27	0.00	0.12	0.40	0.68	0.56	0.74	0.74	0.76	0.74	0.73	0.77	0.98	0.82	0.96	0.97	0.96	0.97	0.96	0.96	0.96	0.96	0.97	0.96	0.97	0.95	0.95	0.97	0.98	0.94	
Laudate dominum	0.52	0.50	0.44	0.68	0.51	0.66	0.28	0.18	0.12	0.00	0.34	0.68	0.58	0.75	0.75	0.78	0.74	0.73	0.78	0.98	0.82	0.96	0.96	0.96	0.97	0.96	0.96	0.96	0.96	0.97	0.96	0.97	0.95	0.95	0.97	0.98	0.94	
De profundis	0.62	0.61	0.56	0.74	0.59	0.73	0.38	0.37	0.40	0.34	0.00	0.69	0.53	0.78	0.78	0.77	0.74	0.73	0.77	0.98	0.83	0.97	0.98	0.97	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.96	0.98	0.99	0.99	0.94		
Qui confidunt	0.69	0.69	0.68	0.76	0.61	0.75	0.66	0.69	0.68	0.68	0.69	0.00	0.73	0.82	0.81	0.82	0.73	0.73	0.78	1.00	0.89	0.98	0.98	0.98	0.98	0.96	0.98	0.98	0.97	0.97	0.96	0.97	0.96	0.98	0.98	0.97		

7. Alignment snapshot

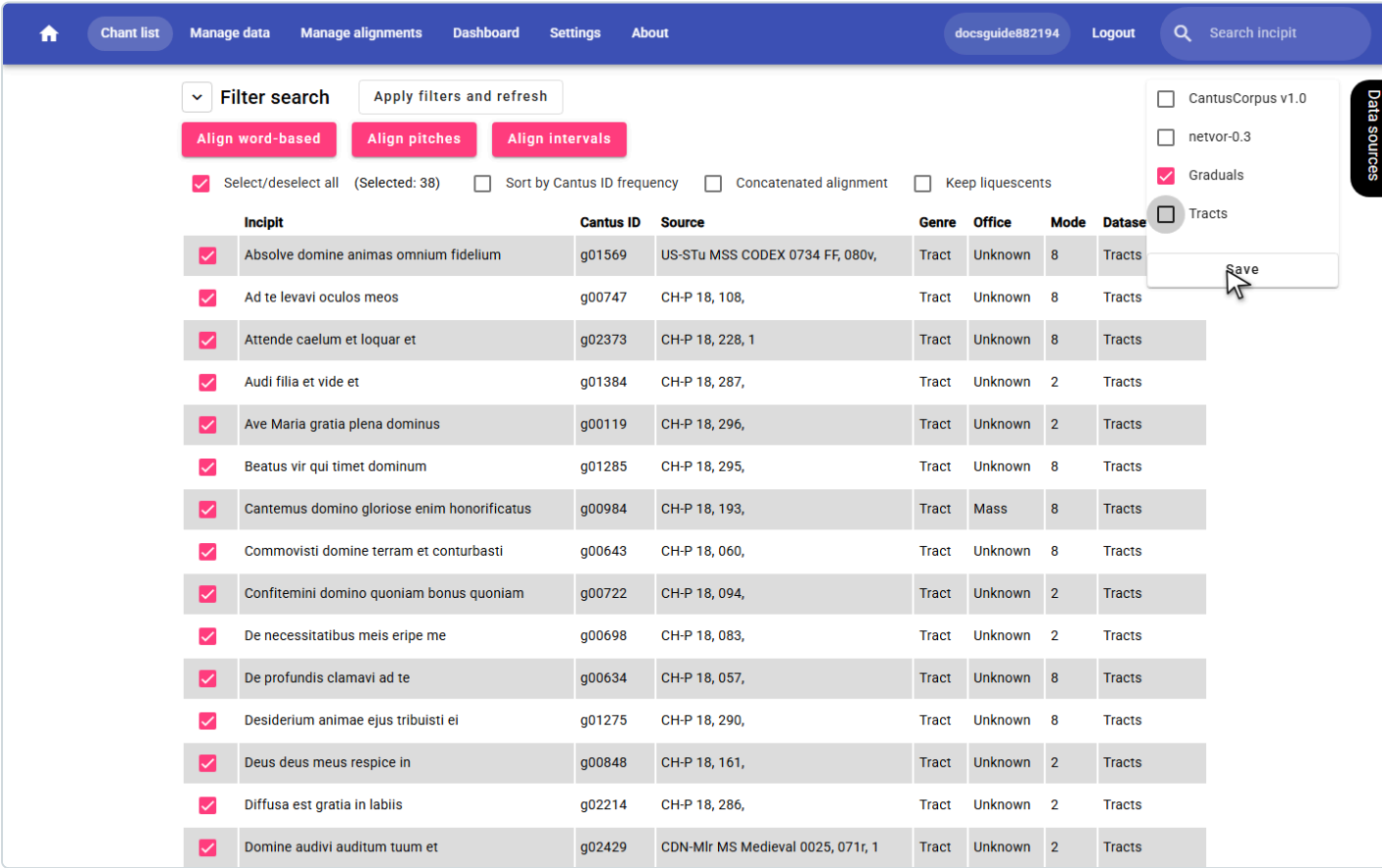
Click **Download Alignment Snapshot PNG** (or **JPEG / PDF**, if that format is selected under **Settings**). The browser downloads a picture of the visible alignment.



Graduals

8. Sort by Cantus ID frequency and pick the top 100

Back on **Chant list**, open **Data sources**, check **Graduals** only, and **Save**.



Check **Sort by Cantus ID frequency**. Chants that share a Cantus ID (the same melody type, in different sources) are grouped, with the most frequent Cantus IDs first.

Home

Chant list

Manage data

Manage alignments

Dashboard

Settings

About

docsguide882194

Logout

Search incipit

Filter search

Apply filters and refresh

Align word-based

Align pitches

Align intervals

☐ Select/deselect all (Selected: 0)

☒ Sort by Cantus ID frequency

☐ Concatenated alignment

☐ Keep liquescents

	Incipit	Cantus ID	Source	Genre	Office	Mode	Dataset name
☐	Christus factus est pro nobis	505005	CH-Fco Ms. 2, 102r,	Gradual	Others		Graduals
☐	Christus factus est pro nobis	505005	CH-P 18, 171,	Gradual	Mass	5	Graduals
☐	Christus factus est pro nobis	505005	D-Mbs Clm 4303, 246r,	Gradual	Day Hours		Graduals
☐	Christus factus est pro nobis	505005	I-Ac 693, 112v,	Gradual	Others		Graduals
☐	Christus factus est pro nobis	505005	I-Ac 693, 112v,	Gradual	Day Hours	*	Graduals
☐	Christus factus est pro nobis	505005	I-Ac 693, 116r,	Gradual	Others	*	Graduals
☐	Christus factus est pro nobis	505005	I-Ac 693, 116r,	Gradual	Others	*	Graduals
☐	Christus factus est pro nobis	505005	I-Ac 693, 119r,	Gradual	Others		Graduals
☐	Haec dies quam fecit [dominus	008414	FIN-Hy F.m.I.26, 010r,	Gradual	Mass	2T	Graduals
☐	Haec dies quam fecit dominus	008414	CH-P 18, 200,	Gradual	Mass	2T	Graduals
☐	Haec dies quam fecit dominus	008414	F-Pn Lat. 15181, 297v,	Gradual	Second Vespers		Graduals
☐	Haec dies quam fecit dominus	008414	I-Ac 693, 121v,	Gradual	Lauds		Graduals
☐	Haec dies quam fecit dominus	008414	V-CVbav B.79, 103r,	Gradual	Matins		Graduals
☐	Haec dies* alleluia	008414	NL-Uu 406 (3 J 7), 097v,	Gradual	None		Graduals
☐	Ab occultis meis munda [me	g00760	US-CApc D-0xpcp, 001v, 3	Gradual	Mass	2	Graduals

Tick the first **100** rows of that sorted list. Do **not** use **Select/deselect all**, which would take every gradual in the dataset.

Then check **Concatenated alignment**. Elements that share a source are joined, so each sequence consists of one source's versions of those melody types laid end to end. This allows you to compare melodies with the same Cantus IDs across sources.

Home

Chant list

Manage data

Manage alignments

Dashboard

Settings

About

docsguide882194

Logout

Search incipit

Filter search

Apply filters and refresh

Align word-based

Align pitches

Align intervals

☐ Select/deselect all (Selected: 100)

☒ Sort by Cantus ID frequency

☒ Concatenated alignment

☐ Keep liquescents

	Incipit	Cantus ID	Source	Genre	Office	Mode	Dataset name
☒	Benedicite dominum omnes angeli ejus	g02712	CH-P 18, 337,	Gradual	Mass	3	Graduals
☒	Benedicta et venerabilis es tu	g03112	NZ-Wt MSR-03, 186r,	Gradual	Mass	4	Graduals
☒	Benedicta et venerabilis es virgo	g01409	CH-P 18, 329,	Gradual	Mass	4	Graduals
☒	Benedictus dominus deus Israel qui	g00614	CH-P 18, 049,	Gradual	Mass	7	Graduals
☒	Benedictus es domine qui intueris	g01117	CH-P 18, 238,	Gradual	Mass	5	Graduals
☒	Benedictus qui venit in nomine	g00548	CH-P 18, 031,	Gradual	Mass	5	Graduals
☒	Bonum est confiteri domino et	g01207	CH-P 18, 105,	Gradual	Mass	5	Graduals
☒	Clamaverunt justi et dominus exaudivit	g01326	CH-P 18, 304,	Gradual	Mass	7	Graduals
☒	Constitues eos principes super omnem	g00006	CH-P 18, 314,	Gradual	Mass	5	Graduals
☒	Custodi me domine ut pupillam	g01179	CH-P 18, 086,	Gradual	Mass	1	Graduals
☒	Deus exaudi orationem meam auribus	g00810	CH-P 18, 142,	Gradual	Mass	8	Graduals
☒	Dirigatur oratio mea sicut incensum	g00713	CH-P 18, 081,	Gradual	Mass	7T	Graduals

9. Align, save, snapshot

Click **Align pitches** and wait until the alignment page is ready.

The screenshot shows the top navigation bar with links: Home, Chant list, Manage data, Manage alignments, Dashboard, Settings, and About. On the right, there's a user profile 'docsguide882194', a 'Logout' button, and a search bar 'Search incipit'. Below the navigation bar, there's a 'Filter search' dropdown and an 'Apply filters and refresh' button. Three buttons are visible: 'Align word-based', 'Align pitches' (highlighted with a mouse cursor), and 'Align intervals'. At the bottom, there are checkboxes for 'Select/deselect all' (Selected: 100), 'Sort by Cantus ID frequency' (checked), 'Concatenated alignment' (checked), and 'Keep liquescents'.

The result has one sequence per source (not 100 separate chants). Each sequence is that source's record of the selected Cantus IDs laid end to end. If a source is missing any Cantus ID, the corresponding segment is left without any notes, allowing you to compare only the melodies for which a record is available.

The screenshot shows the 'Display options' section on the left with checkboxes for 'Show headers', 'Show text', 'Show colors', 'Show conservation', and 'Segment words'. The 'Display mode' section has two tabs: 'Volpiano' (selected) and 'Raw'. Below these are '12 sequences'. On the right, there are buttons for 'Save', 'Download Alignment JSON', 'Download FASTA', 'Download Newick', 'Distance Matrix', 'Download Distance Matrix', 'Guide Tree', and 'Download Alignment Snapshot PNG'. A 'Phylogenetic Analysis' button is also present. The main area displays two musical staves with notes and lyrics: 'di- - es quam fe- cit do- - mi- - ni- es- su- te- - ni- et lae- - te- mur in e- - a'.

Save the alignment, then **Download Alignment Snapshot PNG**.

The screenshot shows the same interface as before, but with a modal dialog box open. The dialog has the title 'Enter alignment name' and a text input field containing 'ncatenated (docs) 1686'. Below the input field is a 'Confirm' button. The background interface is dimmed.

The screenshot shows the top navigation bar with links: Home, Chant list, Manage data, Manage alignments, Dashboard, Settings, and About. On the right, there's a user profile 'docsguide882194', a 'Logout' button, and a search bar labeled 'Search incipit'. Below the navigation bar, the 'Display options' section on the left includes checkboxes for 'Show headers', 'Show text', 'Show colors', 'Show conservation', and 'Segment words'. The 'Display mode' section has buttons for 'Volpiano' (selected) and 'Raw', and 'Notes' (selected) and 'All'. To the right, there are buttons for 'Save', 'Download Alignment JSON', 'Download FASTA', 'Download Newick', 'Distance Matrix', 'Download Distance Matrix', 'Guide Tree', and 'Download Alignment Snapshot PNG'. At the bottom center, a pink button labeled 'Phylogenetic Analysis' is highlighted with a mouse cursor.

10. Phylogenetic analysis

Click **Phylogenetic Analysis**. This needs **at least four** sequences. It runs MrBayes; the number of generations is set under **Settings**.

This is a close-up of the 'Phylogenetic Analysis' button, which is pink and located at the bottom of the display options panel. A mouse cursor is pointing at it. Above it, the 'Download Alignment Snapshot PNG' button is visible. The 'Display mode' section shows 'Notes' selected.

When it succeeds you get:

- the **Newick** string (**Copy** or **Download Newick**)
- **Download nexus.con.tre**, **Download MrBayes script**, **Download nexus alignment**
- an [IcyTree](#) frame: copy the Newick, then in the frame use **File** → **Enter tree directly...** and paste (the in-page **Help** button lists these steps)

The screenshot shows the IcyTree application window. At the top, there's a menu bar with 'File', 'Style', 'Search', 'Statistics', 'Help', and a lightbulb icon. Below the menu bar, the text 'Newick Phylogenetic Tree' is displayed. To the right of this text are buttons for 'Download Newick', 'Download nexus.con.tre', 'Download MrBayes script', and 'Download nexus alignment'. Below the text, there's a text area containing a Newick string:

```
(D_Mbs_Clm_4303]&prob=1.00000000e+00,prob_stdde v=0.00000000e+00,prob_range=
```

. Below the text area are 'Copy' and 'Help' buttons. The main area of the application is a large dashed box containing the text: 'IcyTree', 'Drag and drop a tree file', 'HERE', and 'or load using the File menu'.

11. Manage alignments

Open **Manage alignments**. Saved alignments are listed on the left.

- Select one and click **Show** to reopen the alignment page.
- Select one and click **Delete** to remove it from your account.

The screenshot shows the 'Manage alignments' interface. On the left, under 'Manage alignments', there is a list of two items: 'Graduals concatenated (docs) 1686' (selected with a red radio button) and 'Tracts pitch alignment (docs) 1686' (unselected with a grey radio button). Below this list are 'Show' and 'Delete' buttons. The main content area is titled 'Upload alignment' and contains an 'Alignment name' input field, a 'Choose File' button (labeled 'No file chosen'), an 'Upload' button, and a 'Help: Alignment Format' dropdown menu.

On the right, **Upload alignment** restores a file you previously downloaded with **Download Alignment JSON** on the alignment page. That file is a ChantLab alignment JSON, not a chant CSV.

1. Type an **Alignment name**.
2. Choose the `.json` file.
3. Click **Upload**.

The new name then appears in the list on the left, and you can **Show** it like any other saved alignment.

This screenshot is identical to the previous one, but with a mouse cursor clicking the 'Upload' button in the 'Upload alignment' section.